# Reminders and announcements

- During lectures you can submit questions to: vevox.app ID: 107-018-952
- You can find answers to questions in the COMP1870 Teams Channel (link in Minerva) (along with other discussion).
- Maths Lunch Hours, Mondays and Wednesdays, 1.00pm - 2.00pm. Bragg GR.18.
- My student hour: Tuesdays 1.00pm - 2.00pm, Bragg building foyer.

# Reminders and announcements

Starting on Monday 2 November, COMP1870 lectures change to:

- Mondays from 4.00pm to 6.00pm in Mechanical Engineering Lecture Theatre A (1.33).
- Thursdays from 1.00pm to 3.00pm in Chemical and Process Engineering Lecture Theatre A (2.23)

# Revision

- A **relation** from $A$ to $B$ is any subset $R \subseteq A \times B$.
- We write $aRb$ to mean that $(a, b)$ is in $R$.
- If $A = B$, then $R$ is a **relation on** $A$.

| **Property** | **Definition** | **Characterisation via arrows** |
|---|---|---|
| Reflexive | $(a, a) \in R$ for all $a \in A$ | Every element has a loop |
| Symmetric | $(a, b) \in R$ implies $(b, a) \in R$ | Every arrow is two-way |
| Anti-symmetric | $(a, b) \in R$ and $(b, a) \in R$ implies $a = b$ | The only two-way arrows are loops |
| Transitive | $(a, b) \in R$ and $(b, c) \in R$ implies $(a, c) \in R$ | If we can get from $a$ to $c$ in two steps, we can get there in one step |

# Revision

Let $A$ be a set.

- An **equivalence relation** on $A$ is a relation on $A$ that is reflexive, symmetric, and transitive.
- If $R$ is an equivalence relation on $A$ and $a$ is an element of $A$, then $[a] = \{x \in A : aRx\}$ is the **equivalence class** containing $a$.
- The equivalence classes **partition** $A$. (This means that each element of $A$ is in exactly one equivalence class.)

# Partial orders

**Definition 7.11.** Let $A$ be a set and let $R \subseteq A \times A$ be a relation on $A$. If $R$ is reflexive, anti-symmetric, and transitive, then $R$ is a **partial order**.

**Proposition 7.12.** Let $A$ be a set and let $R \subseteq \mathcal{P}(\mathcal{A}) \times \mathcal{P}(\mathcal{A})$ be the relation on subsets of $A$ such that $(X, Y)$ is in $R$ if and only if $X$ is a subset of $Y$. Then $R$ is a partial order.

# Hasse diagrams

Partial orders can be represented by diagrams called **Hasse diagrams**.

A Hasse diagram can be obtained from the directed graph diagram by the following steps.

- Since the relation is reflexive, we know that every element has a loop. Therefore we can omit loops.
- Since the relation is anti-symmetric it cannot contain any **directed cycles** (Proposition 7.13). So we can draw the directed graph with every arrow directed upwards on the page. Therefore we can omit arrowheads.
- Since the relation is transitive, we can omit any arrows implied by transitivity.

# Hasse diagrams

# Cardinality

The number of elements in a set is called the **cardinality** or the **size** of the set.

If $A$ is a set, then the cardinality of $A$ is denoted by $|A|$. (Alternative notation: $\#A$.)

# Addition rule

Let $A$ and $B$ be sets. If $A \cap B$ is empty, then we say that $A$ and $B$ are **disjoint**.

**Proposition 8.3.** Let $A$ and $B$ be disjoint finite sets. Then

$$|A \cup B| =$$

This is known as the **addition rule** (or addition principle). Note that it does not hold if $A \cap B$ is non-empty.

Say that $A_1, A_2, \ldots, A_m$ are sets. If each pair of these sets is disjoint, then we say they are **pairwise disjoint**. In this case,

$$|A_1 \cup A_2 \cup \cdots \cup A_m| =$$

# Multiplication rule

Let $A$ and $B$ be sets. Then $A \times B = \{(a, b) \mid a \in A, b \in B\}$ is the **Cartesian product** of $A$ and $B$.

**Proposition 8.4.** Let $A$ and $B$ be finite sets. Then

$$|A \times B| =$$

This is known as the **multiplication rule** (or principle).

Say that $A_1, A_2, \ldots, A_m$ are sets. Then $A_1 \times A_2 \times \cdots \times A_m$ is the set of all tuples $(a_1, a_2, \ldots, a_m)$ such that $a_i$ is an element of $A_i$ for each possible $i$. Then

$$|A_1 \times A_2 \times \cdots \times A_m| =$$

# Multiplication rule

The multiplication rule says:

> *Suppose we have $n$ choices to make. Assume that for the first choice we can choose from $i_1$ options, for the second choice we can choose from $i_2$ options, and so on, so that for choice number $j$, we have $i_j$ options. The total number of ways we can make the choices is*
>
> $$i_1 \times i_2 \times \cdots \times i_n$$

# Applications of the multiplication rule

Remember that when $A$ is a set, $\mathcal{P}(A)$ is the **power set** of $A$: the set of all subsets of $A$.

**Proposition 8.5.** Let $A$ be a finite set such that $|A| = n$. Then $|\mathcal{P}(A)| =$

# Permutations

Let $A$ be a set. A **permutation** of $A$ is an ordering of the elements of $A$. When $n$ is a non-negative integer, **$n$ factorial** is

$$n! = n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1.$$

It is convenient to define 0! to be 1.

**Proposition 8.6.** Let $A$ be a set with $|A| = n$. The number of permutations of $A$ is $n!$.

# Ordered selections

Let $A$ be a set. When we make an **ordered selection** from $A$, we choose elements from $A$, and the order in which we choose them is **significant**.

**Example.** Say that $A = \{1, 2, 3, 4, 5\}$. In an ordered selection, choosing 3 first and 4 second is different from choosing 4 first and 3 second.

# Ordered selections

**Proposition 8.7.** Let $A$ be a set with $|A| = n$. The number of ways we can make an ordered selection of $k$ distinct elements of $A$ is

$$n \cdot (n-1) \cdot (n-2) \cdots (n-k+1) = \frac{n!}{(n-k)!}.$$

# Unordered selections

Let $A$ be a set. When we make an **unordered selection** from $A$, we choose elements from $A$, and the order in which we choose them is **not significant**.

**Example.** Say that $A = \{1, 2, 3, 4, 5\}$. In an ordered selection, choosing 3 first and 4 second is the same as choosing 4 first and 3 second.

# Binomial coefficients

**Proposition 8.8.** Let $A$ be a set with $|A| = n$. The number of ways we can make an unordered selection of $k$ distinct elements of $A$ is

$$\frac{n!}{(n-k)!k!}.$$

# Binomial coefficients

**Proposition 8.8.** Let $A$ be a set with $|A| = n$. The number of ways we can make an unordered selection of $k$ distinct elements of $A$ is

$$\frac{n!}{(n-k)!k!}.$$

This is equal to the number of subsets of $A$ with size $k$.

The number $\dfrac{n!}{(n-k)!k!}$ is denoted by $\binom{n}{k}$ and is called "$n$ **choose** $k$". These numbers are called **binomial coefficients**.

# The binomial theorem

$$
\begin{aligned}
(a+b)^0 &= 1 \\
(a+b)^1 &= 1a + 1b \\
(a+b)^2 &= 1a^2 + 2ab + 1b^2 \\
(a+b)^3 &= 1a^3 + 3a^2b + 3ab^2 + 1b^3 \\
(a+b)^4 &= 1a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + 1b^4 \\
(a+b)^5 &= 1a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + 1b^5
\end{aligned}
$$

# The binomial theorem

$$
\begin{aligned}
(a+b)^0 &= 1 \\
(a+b)^1 &= 1a + 1b \\
(a+b)^2 &= 1a^2 + 2ab + 1b^2 \\
(a+b)^3 &= 1a^3 + 3a^2b + 3ab^2 + 1b^3 \\
(a+b)^4 &= 1a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + 1b^4 \\
(a+b)^5 &= 1a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + 1b^5
\end{aligned}
$$

$$
\begin{aligned}
(a+b)^0 &= \binom{0}{0} \\
(a+b)^1 &= \binom{1}{0}a + \binom{1}{1}b \\
(a+b)^2 &= \binom{2}{0}a^2 + \binom{2}{1}ab + \binom{2}{2}b^2 \\
(a+b)^3 &= \binom{3}{0}a^3 + \binom{3}{1}a^2b + \binom{3}{2}ab^2 + \binom{3}{3}b^3 \\
(a+b)^4 &= \binom{4}{0}a^4 + \binom{4}{1}a^3b + \binom{4}{2}a^2b^2 + \binom{4}{3}ab^3 + \binom{4}{4}b^4 \\
(a+b)^5 &= \binom{5}{0}a^5 + \binom{5}{1}a^4b + \binom{5}{2}a^3b^2 + \binom{5}{3}a^2b^3 + \binom{5}{4}ab^4 + \binom{5}{5}b^5
\end{aligned}
$$

# The binomial theorem

**Theorem 8.9.** Let $n$ be a non-negative integer. Then

$$\begin{aligned}(a+b)^n &= \binom{n}{0}a^n b^0 + \binom{n}{1}a^{n-1}b^1 + \cdots \\ &\qquad + \binom{n}{n-1}a^1 b^{n-1} + \binom{n}{n}a^0 b^n \\ &= \sum_{i=0}^{n}\binom{n}{i}a^{n-i}b^i\end{aligned}$$

# Pascal's Triangle

This array is called **Pascal's Triangle**.

$$
\begin{array}{ccccccccccc}
 & & & & & \binom{0}{0} & & & & & \\
 & & & & \binom{1}{0} & & \binom{1}{1} & & & & \\
 & & & \binom{2}{0} & & \binom{2}{1} & & \binom{2}{2} & & & \\
 & & \binom{3}{0} & & \binom{3}{1} & & \binom{3}{2} & & \binom{3}{3} & & \\
 & \binom{4}{0} & & \binom{4}{1} & & \binom{4}{2} & & \binom{4}{3} & & \binom{4}{4} & \\
\binom{5}{0} & & \binom{5}{1} & & \binom{5}{2} & & \binom{5}{3} & & \binom{5}{4} & & \binom{5}{5}
\end{array}
$$

$$
\begin{array}{ccccccccccc}
 & & & & & 1 & & & & & \\
 & & & & 1 & & 1 & & & & \\
 & & & 1 & & 2 & & 1 & & & \\
 & & 1 & & 3 & & 3 & & 1 & & \\
 & 1 & & 4 & & 6 & & 4 & & 1 & \\
1 & & 5 & & 10 & & 10 & & 5 & & 1
\end{array}
$$

# Pascal's Identity

**Proposition 8.10.** Let $n$ and $k$ be positive integers where $k < n$. Then

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

# The pigeonhole principle

Imagine that we distribute $n$ pigeons into $k$ pigeonholes. There may be many such distributions.

We examine these distributions and look at the maximum number of pigeons in a pigeonhole.

Can we give a lower bound on this maximum?

# The pigeonhole principle

**Definition 8.11.** Let $x$ be a real number.

- $\lfloor x \rfloor$ is the largest integer that is less than or equal to $x$.
  ($\lfloor x \rfloor$ is called the **floor** of $a$.)
- $\lceil x \rceil$ is the smallest integer that is greater than or equal to $x$.
  ($\lceil x \rceil$ is called the **ceiling** of $x$.)

# The pigeonhole principle

**Proposition 8.12.** If $n$ pigeons are distributed into $k$ pigeonholes, the maximum number of pigeons in the same pigeonhole is at least $\lceil n/k \rceil$.

# The pigeonhole principle

**Proposition 8.13.** Let $f$ be a function from $A$ to $B$, where $|A| = n$ and $|B| = k$. Then there is an element $b \in B$ such that at least $\lceil n/k \rceil$ elements in $A$ are mapped to $b$ by $f$.

# The pigeonhole principle

**Example.** Let $D$ be a set of 2048 data values. Assume that $f : D \to H$ is a hash function where $H$ is a set of 40 hash values.

There is some hash value $h \in H$ such that there are at least _____ data values in $D$ that are mapped to $h$.